Darren Lee
danalee@ucsc.edu
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CSE 13S Spring 2021
Assignment 2: A Small Numerical Library
Design Document

New changes in red

Purpose

The purpose of this assignment is to create a small numerical library and test harness. The library will include arcSin, arcCos, arcTan, and Log functions. I will use the trigonometric identities, binary search, and Newton's method for arcSin and trigonometric identities for arcCos and arcTan. I will be using Newton's method to approximate the logarithm function. The test harness will allow me to compare my results to the standard <math.h> library and create a table showing the values and differences. ArcSin and arcCos will be tested in the range [−1,1] with steps of 0.1. ArcTan and Log will be tested in the range[1,10] with steps of 0.1.

Layout/Structure

- The code for the functions will be in mathlib.c. This will contain the arcSin, arcCos, arcTan, and Log functions.
- Mathlib-test.c contains the main() program and acts as a test harness for the math library.
- The getopt() command line options that mathlib-test.c supports are:
    - -a: to run all tests.
    - -s: to run arcsin tests.
    - -c: to run arccos tests.
    - -t: to run arctan tests.
    - -l: to run log tests.

Functionality

- The arcSin function will be implemented using the inverse method with Newton's method and binary search. Values close to -1 will be calculated with binary search and other values less than 1 will be calculated with Newton's method. I will use Newton's method and the identity $\cos(\arcsin(x)) = \sqrt{1-x^2}$ to find arcSin using arcCos(Sqrt(1-x^2)) for values close to 1.

- The arcCos function will be implemented using the arcSin function. I will use the equation $$\arccos(x) = \frac{\pi}{2} - \arcsin(x).$$
- The arcTan function will be implemented using the arcCos function and the equation

$$\arctan(x) = \arccos\left(\frac{1}{\sqrt{x^2+1}}\right), \quad x > 0.$$

- The Log function will be implemented using Newton's method by computing the inverse of e^x. The following equation will allow us to implement logarithms with base e:

$$x_{k+1} = x_k + \frac{y - e^{x_k}}{e^{x_k}}.$$

Pseudocode for mathlib.c

```
Abs(x) function: (from Piazza)
        Return (x < 0) ? -x : x

Exp(x) function: (from Piazza)
        term = 1, sum = 1
        For (k=1; Abs(term) > EPSILON; k++):
                term *= x/k
                sum += term
        Return sum

Sqrt(x) function: (from Piazza)
        y = 1.0
        assert (x >= 0)
        For (guess = 0.0; Abs(y-guess) > EPSILON; y = (y + x / y) / 2.0):
                Guess = y
        Return y
```

Removed Pow function because it was unnecessary

```
arcSin(x) function:
        k = 0.0
        prev = sin(k)
        j = 1.0
        p = cos(k)
        If (x>0.99):
                While (Abs(p - Sqrt(1 - x * x)) > EPSILON):
                        j = j - (Sqrt(1 - x * x) - p) / sin(j)
```

```
                p = cos(j)
            Return j

        If (x < -0.99):
            Declare m, l = -π/2, h = π/2
            Do:
                m=(l + h) / 2.0
                If (Sin(m) < x):
                    l = m
                Else:
                    h = m
            While (Abs(l - h) > EPSILON / 2)
            Return m

        Else:
            while(Abs(prev - x) > EPSILON)
                k = k + (x - prev) / cos(k)
                prev = sin(k)
            Return k

arcCos(x) function:
    Return π / 2 - arcSin(x)

arcTan(x) function:
    Return arcCos(1 / Sqrt(x * x + 1))

Log(x) function:
    Declare guess
    k = 1.0
    For (guess = 0.0; Abs(k - guess) > EPSILON; k = k + (x - Exp(k)) / Exp(k)):
        guess = k
    Return k
```

Pseudocode for mathlib-test.c

```
Define OPTIONS “asctl”
print_arcSin() function:
    Print “ x        arcSin        Library        Difference\n"
    For (i = -1.0; i <= 1.0; i += 0.1):
        Print “%7.4lf%16.8lf%16.8lf%16.10lf\n", i, arcSin(i), asin(i), arcSin(i) - asin(i)
    Return
```

```
Print_arcCos() function:
        Print " x        arcCos        Library        Difference\n"
        For (i = -1.0; i <= 1.0; i += 0.1):
                Print "%7.4lf%16.8lf%16.8lf%16.10lf\n", i, arcCos(i), acos(i), arcCos(i) - acos(i)
        Return

print_arcTan() function:
        Print " x        arcTan        Library        Difference\n"
        for (i = 1.0; i <= 10.0; i += 0.1):
        Print "%7.4lf%16.8lf%16.8lf%16.10lf\n", i, arcTan(i), atan(i), arcTan(i) - atan(i)
        Return

print_Log() function:
        Print " x        Log           Library        Difference\n"
        for (double i = 1.0; i <= 10.0; i += 0.1):
        Print "%7.4lf%16.8lf%16.8lf%16.10lf\n", i, Log(i), log(i), Log(i) - log(i)
        Return

Main (argc, **argv) function:
        opt=0, sin_test=0, cos_test=0, tan_test=0, log_test=0
        While ((opt=getopt(argc, argv, OPTIONS) != -1):
                switch(opt):
                Case 'a':
                        if (!sin_test):
                                print_arcSin()
                                sin_test = 1

                        if (!cos_test):
                                print_arcCos()
                                cos_test = 1

                        if (!tan_test):
                                print_arcTan()
                                tan_test = 1

                        if (!log_test):
                                print_Log();
                                log_test = 1

                        break
                case 's':
                        if (!sin_test):
                                print_arcSin()
                                sin_test = 1
```

```
                break
        case 'c':
                if (!cos_test):
                        print_arcCos()
                        cos_test = 1
                Break
        Case ‘t’:
                if (!tan_test):
                        print_arcTan()
                        tan_test = 1
                Break
        Case ‘l’
                if (!log_test):
                        print_Log();
                        log_test = 1
                break
if (!sin_test && !cos_test && !tan_test && !log_test):
        Print "Program usage ./mathlib-test -[asctl]\n"
        Print "  -a   Runs all tests (arcsin, arccos, arctan, log)\n"
        Print “  -s   Runs arcsin tests\n"
        Print "  -c   Runs arccos tests\n"
        Print "  -t   Runs arctan tests\n"
        Print "  -l   Runs log tests\n"
        return 0
```

Draft Work

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

$$\sin^{-1}(y) = x$$

$$\sin(\sin^{-1}(y)) = \sin x$$

$$y = \sin x$$

$$y - \sin x = 0$$

$$x_{k+1} = x_k - \frac{y - \sin x_k}{-\cos x_k}$$

$$x_{k+1} = x_k + \frac{y - \sin x_k}{\cos x_k}$$